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Enformer: An encoder-decoder generative model to enhance prediction of disease outcomes using electronic health records

Zhichao Yang
University of Massachusetts Amherst
Avijit Mitra
University of Massachusetts Amherst
Weisong Liu
University of Massachusetts Lowell
Dan Berlowitz
University of Massachusetts Lowell
Hong Yu (Hong_Yu@uml.edu)
University of Massachusetts Lowell

Article

Keywords:

Posted Date: June 15th, 2023

DOI: https://doi.org/10.21203/rs.3.rs-2922823/v1

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Additional Declarations: There is **NO** Competing Interest.

Version of Record: A version of this preprint was published at Nature Communications on November 29th, 2023. See the published version at https://doi.org/10.1038/s41467-023-43715-z.

Abstract

Deep learning transformer-based models using longitudinal electronic health records (EHRs) have shown a great success in prediction of clinical diseases or outcomes. Pretraining on a large dataset can help such models map the input space better and boost their performance on relevant tasks through finetuning with limited data. In this study, we present Enformer, a generative encoder-decoder model with transformer that was pretrained using a new pretraining objective - predicting all diseases and outcomes of a patient at a future visit from previous visits. Enformer's encoder-decoder framework, paired with the novel pretraining objective, helped it achieve the new state-of-the-art (SOTA) performance on multiple clinical prediction tasks. Comparing with the previous SOTA model, Enformer improved area under the precision–recall curve (AUPRC) by 2% ($p<0.001$) for pancreatic cancer onset and by 24% ($p<0.001$) for intentional self-harm in patients with PTSD. The high performance in predicting intentional self-harm shows the potential of Enformer in building effective clinical intervention systems. Enformer is also generalizable and can be easily finetuned for clinical prediction tasks with limited data.

Introduction

The widespread adoption of electronic health records (EHRs) across US hospitals has led to the development and adoption of numerous data mining and statistical techniques for EHRs. Longitudinal EHRs have been successfully used to predict clinical diseases or outcomes.[1–3] Early work applied regression and traditional machine learning (ML) based models (e.g., support vectors machines, random forest, and gradient boosting) to predict single disease or outcome. Examples include congestive heart failure,[4] sepsis mortality,[5] mechanical ventilation,[5] septic shock,[6] type 2 diabetes,[7] and development of post-traumatic stress disorder (PTSD)[8], among others.

With the availability of large cohorts and computational resources, deep learning based models can outperform traditional ML models[9–15]. State-of-the-art (SOTA) models in EHR-based predictive modeling achieved this through the pretrain-finetune paradigm - a two-step process where the model is first trained on large-scale longitudinal EHRs to learn the representations of clinical features such as, International Classification of Diseases (ICD) codes (pretrain) and then further trained to adapt to specific tasks e.g., outcome prediction (finetune). Models such as Med-BERT[12], BEHRT[13], and BRLTM[14] fall in this category. However, their pretraining objectives were limited in predicting a fraction of ICD codes within each visit. In reality, most patients have multiple diseases or outcomes at once,[16] many of which are highly correlated (such as obesity, diabetes, and hypertension[17–19]) and thus collectively contribute to the disease or outcome trajectories. Therefore, a novel pretraining strategy, which predicts the complete set of diseases and outcomes within a visit, might improve clinical predictive modeling.

In this study, we propose Enformer, an innovative denoising sequence to sequence transformer[20] model that was pretrained on 6.5 million patients' EHRs to predict complete ICD codes of a visit. Enformer can be further finetuned for single disease or outcome predictions. Unlike previous EHR-based models[12–15] which rely on the bidirectional (left-to-right and right-to-left) encoder representation from transformers (BERT) framework,[21] Enformer used a transformer-based encoder-decoder generative framework to predict future ICD codes during pretraining. The unidirectional (left-to-right) decoder in such an encoder-decoder framework is more similar to the use case of future disease or outcome predictions based on history of past diseases or outcomes (past-to-future) compared to the bidirectional encoder-only framework.

Although the encoder-decoder framework was originally designed to generate next sentence given previous sentences as context,[22,23] we repurposed the framework for Enformer to generate the ICD codes of the next visit given previous EHRs (context). Enformer can utilize cross-attention[20] by identifying relevant ICD codes from previous visits to predict future ICD codes. The decoder then predicts ICD codes one after another by using already predicted diagnostic ICD codes to predict next ICD codes. Furthermore, Enformer includes date of each visit to integrate temporal information, whereas previous transformer-based predictive models only included their sequential order.[12–15] Specific date of each visit is an important feature in predictive modeling as importance of predictor in a visit can vary over time.[1,24,25]

We evaluated Enformer for a broad range of disease and outcome predictions. In addition to predictions of ICD codes, we evaluated Enformer on two challenging and clinically important disease and outcome prediction tasks: pancreatic cancer prediction and intentional self-harm prediction among PTSD patients. In summary, our key contributions are as follows:

1. We propose a new pretraining objective that predicts all diseases or outcomes of a future visit using longitudinal information from the previous visits. Such pretraining objective can leverage the clinical relations among diseases and outcomes.
2. This is the first study that explores a generative encoder-decoder framework to predict patients' ICD codes using their longitudinal EHRs. Our encoder-decoder framework outperformed the encoder-based models in part due to the decoder self-attention and cross-attention mechanisms. Enformer outperformed SOTA BERT models on both common and uncommon ICD code predictions. In particular, the improvements for uncommon were substantial.
3. Enformer achieved a positive predictive value (PPV) of 8.45% for prediction of intentional self-harm among the top 10% PTSD patients at high predicted risk. A recent study has shown that a practical suicide prevention tool must have a achieve above 1.7% PPV to minimize the resource cost and intrusion directed at patients who will never attempt self-harm.[26] This shows the potential of deploying Enformer for clinical screening and interventions.
4. We validated the generalizability of Enformer using both internal and external datasets. Internally, we evaluated Enformer on "never-seen" in-domain EHR data from VHA facilities. Externally, we evaluated Enformer on out-of-domain data from a non-VHA hospital. Our results demonstrated a strong transfer learning[27]capability of Enformer, which can greatly benefit hospitals with limited data and computing resources.

Results

Data

Our pretraining cohort comprises 6,475,218 patients who received care from more than 1,200 health care facilities of the US Veterans Health Administrations (VHA) from 1/1/2016 to 12/31/2019. To evaluate pretrained models, we created two disease/outcome agnostic prediction (DOAP) datasets – one for common and one for uncommon diseases/outcomes. We selected 10 ICD-10 codes with the highest prevalence (prevalence ratio > 2%) in our pretraining cohort for our common disease/outcome DOAP dataset. As for the set of uncommon diseases/outcomes, we followed the FDA guidelines[28] to randomly select 10 ICD-10 codes with a prevalence ratio ranging from 0.04% to 0.05% in our pretraining cohort. The lists of common and uncommon diseases/outcomes are shown in Table 1.

To evaluate finetuned models for single disease and outcome predictions with low prevalence ratio, we created two EHR datasets for two important prediction tasks: pancreatic cancer (single disease) and intentional self-harm among patients with PTSD (single outcome). Early screening, early diagnosis and early treatment of pancreatic cancer are critical for this deadly disease.[29,30] Med-BERT[12] has been used to predict pancreatic cancer. Intentional self-harm is common among the US military Veterans with PTSD.[31] The detailed pretraining and finetuning cohort definitions are presented in Methods section. We compare these cohorts in Supplementary Table 1.

To assess the generalizability, we evaluated Enformer on both "unseen" in-domain EHR data and out-of-domain non-VHA EHR data. The unseen in-domain EHR data were collected from VHA facilities not included in the pretraining cohort and we created a dataset for the intentional self-harm prediction task among PTSD patients. As the out-of-domain non-VHA EHR data, we used the MIMIC-IV dataset[32] to build a non-VHA DOAP dataset. The MIMIC-IV dataset includes intensive care unit patients admitted to the Beth Israel Deaconess Medical Center in Boston, Massachusetts. Since the dataset contains information from 2008 to 2019 but the implementation of ICD-10 started from October 2015, we only selected patients with the ICD-10 records to match our implementation for the cohorts from VHA, resulting in a dataset of 29,482 patients.

Longitudinal EHRs

As shown in Figure 1, Enformer takes longitudinal EHRs as input. To compare Enformer with the previous SOTA EHR-based models using BERT,[12,14] we included demographic information and ICD-10 codes as predictors. Demographic information includes gender, age, race, and marital status. The attributes (e.g., male) of each category (e.g., gender) were appended as individual predictors. Following previous work,[12] we grouped ICD codes at visit level. Within a visit, we ordered ICD codes from high to low priority, as assigned by health care providers, where the primary diagnosis is typically given the highest priority, followed by the secondary diagnosis and so on.

Multiple visits of each patient formed a time-stamped (by date of visit) input of a sequence of ICD-10 code groups (Figure 2). We used multi-level embeddings.[12] Embeddings are trainable fixed-dimensional vectors that were used to represent predictors in a continuous vector space and were learned during the pretraining process (details in the next section). We represented each visit in an input sequence using a visit embedding and each ICD code using a code embedding. To embed the specific date of each visit, we applied sinusoidal position embedding[20] to the numerical format of date. Finally, each input embedding was constructed by summing up the corresponding code embedding, visit embedding, and date embedding (Figure 2a).

Pretrain-Finetune Paradigm

With longitudinal EHRs as input, we first pretrained models on the pretraining cohort of 6,475,218 patients and then finetuned the model for single disease or outcome prediction, as shown in Figure 3. During the pretraining, the model was trained to recover the original longitudinal EHRs from corrupted (masked) longitudinal EHRs.

Previous EHR-based BERT models corrupted longitudinal EHRs by randomly sampling 25% ICD codes and replacing them with mask (code masking).[12–14] Enformer, on the other hand, masked all ICD codes in a single visit (visit masking). A comparative example is illustrated in Figure 2b. In other words, Enformer was pretrained to predict the complete set of ICD codes of a patient's future visit, given demographic information and longitudinal ICD codes up to the current visit.

Enformer Architecture

Enformer uses an encoder–decoder transformer architecture.[22,23] The encoder processes the input embeddings and generates a set of hidden representations for each predictor. Unlike the encoder-only transformer architecture used by existing EHR-based BERT models,[12–14] Enformer performs cross-attention over the hidden representations from the encoder and assigns an attention weight for each representation. These weighted representations are then fed to the decoder, which generates ICD codes of the future visit as illustrated in Figure 3. The decoder generates ICD codes following the sequential order of code priority within a visit. In other words, it first generates primary diagnosis, and then generates secondary diagnosis based on primary diagnosis. This step is repeated until all diagnoses of a future visit are completed. Our results showed that the Enformer architecture outperformed the encoder-only architecture.

Evaluation Metrics

We used Jaccard similarity score to measure model performance on the DOAP task. Jaccard similarity score measures the similarity between ground truth and model predictions as the size of the intersection of ground truth codes and the prediction codes divided by the size of their union. A higher score indicates better model performance. We also reported positive predictive value (PPV or precision), area under the receiver operating characteristic curve (AUROC), and area under the precision recall curve (AUPRC) to measure models' performance on single disease or outcome predictions. PPV is the fraction of true positives from all predicted positives. AUROC is the area under sensitivity and false positive rate curve. Sensitivity (or recall) is the number of true positives divided by the number of ground truth positives. False positive rate is the number of false positives divided by total number of negatives. AUPRC is the area under PPV and sensitivity curve, and it has shown to be an effective measure for highly imbalanced binary classification problems, which include self-harm prediction.[33,34]

We compared Enformer with four strong baseline models: copy, logistic regression, long short-term memory (LSTM),[35] and BERT[21] pretrained on VHA cohort. The copy baseline is essentially making predictions by simply copying the ICD codes from the previous visit. Although simple, this is a strong baseline since it is well known that physicians copy and paste note content.[36,37]

Pretraining Evaluation: Disease or Outcome Agnostic Prediction

DOAP is the task of predicting the ICD codes of a patient's future visit based on patient's demographic information and longitudinal ICD codes up to the current visit. Supplementary Table 2 showed that pretraining substantially improved predictions of both common

and uncommon diseases/outcomes. Compared to the copy baseline, Enformer improved the Jaccard similarity scores for recurrent common diseases/outcomes by 64% (average 30.41, from 45.49 to 74.85) and for uncommon diseases/outcomes by 83% (average 22.04, from 26.54 to 48.58). Compared to BERT, Enformer also improved performance by 37% in Jaccard similarity in all prediction subtasks, regardless of disease/outcome category (common or uncommon) and occurrence type (new or recurrent). We found a 15% improvement in Jaccard similarity score on common disease/outcome predictions and 45% improvement on uncommon disease/outcome predictions.

Our Enformer contains three unique components compared to previous medical BERT-based models: 1) visit masking, 2) encoder-decoder architecture, and 3) date embedding. We performed ablation analysis to evaluate the effectiveness of each component. First, we compared an encoder-decoder model that uses visit masking, i.e., masking all ICD codes of a visit, to another encoder-decoder model that uses code masking, i.e., masking a randomly selected fragment of ICD codes of a visit. Our findings revealed that the visit masking performed better (showing an average improvement of 23% in Jaccard similarity score) compared to code masking for all diseases/outcomes tested as shown in Supplementary Table 3. These results demonstrated that pretraining for prediction of all diseases and outcomes outperform traditional pretraining objectives.

Next, we compared encoder-decoder model with encoder-only model (BERT) on DOAP. Our findings showed that the encoder-decoder model outperformed the encoder-only model (with an average improvement of 9% in Jaccard similarity score) across all diseases/outcomes tested (Supplementary Table 2 and 3).

Finally, we conducted an experiment by excluding date embedding from the implementation of Enformer. Data embedding allows Enformer to capture the temporal information of prior visits. Our results indicated that Enformer with date embedding exhibits a moderate improvement (with an average increase of 1% in Jaccard similarity score) than Enformer without date embedding across most diseases/outcomes (Supplementary Table 3).

Finetuning Evaluation: Single Disease or Outcome Prediction

Pancreatic Cancer

Results of pancreatic cancer onset prediction are shown in Table 2. Enformer achieved AUPRC of 82.73 (95% CI: 81.81, 83.65), outperforming both the logistic regression model (63.81; 95% CI: 61.47, 66.15, p<0.001), LSTM (78.46; 95% CI: 77.60, 79.32, p<0.001) and BERT (81.11; 95% CI: 80.53, 81.69, P=0.01).

Intentional Self-Harm in Patients with PTSD

This task is challenging because of its low prevalence (1.9%). As shown in Table 3, AUPRC of Enformer was 16.38 (95% CI, 15.25, 17.52), outperforming BERT (13.17; 95% CI: 11.72, 14.61), LSTM (8.32; 95% CI: 7.02, 9.61), and logistic regression (2.8; 95% CI: 0.49, 5.13) by 24.3%, 55.4%, 96.9%, and 482.9% respectively. We also calculated sensitivity and PPV for a variety of thresholds. As shown in Supplementary Table 4, The PPV of Enformer ranges from 3.11 to 8.45 for 10% to 60% threshold.

Subgroup analyses among different demographics:

The results of the subgroup analyses are shown in Supplementary Table 5. AUPRC was consistent among different genders, ages, races, and marital status. For example, the AUPRC of patients who were more than 80-year-old (4.86% of the cohort) was 17.97 (95% CI, 16.43, 19.51), which was not statistically different from the AUPRC of patients whose ages range from 30 to 39 (19.58% of the cohort): 16.29 (95% CI, 14.41, 18.17).

Effect of historical EHR length on performance:

To examine the impact of how patients' prior history impacts model predictions, we conducted two experiments - "self-harm with short history" (included only the 5 most recent visits) and "self-harm" included all visits). As shown in Table 3, The AUPRC of Enformer using only the 5 prior visits was 13.82. Using all visits, the AUPRC improved by 18% to 16.38.

Generalizability Evaluation

We validated the generalizability of Enformer both internally and externally. Out of 1,239 VHA health care facilities, we found that EHR data from 121 facilities were not included in the pretraining cohort. We created an internal generalizability evaluation dataset using this subset of EHR data for the intentional self-harm prediction task among PTSD patients. Upon evaluating Enformer on this dataset, our results showed no statistical difference in AUPRC between data from unseen facilities (1st bar on the left of Supplementary Figure 1) and data from facilities where at least some of the data were included in the pretraining cohort (other bars of Supplementary Figure 1).

To evaluate Enformer's generalizability on the external dataset, we further finetuned the pretrained Enformer and BERT on the training set of a non-VHA DOAP dataset and evaluated them on the testing set. Enformer outperformed BERT by 8% with a Jaccard similarity score of 24.35, whereas BERT obtained a score of 22.51.

Discussion

In this study we introduced Enformer, a novel generative deep neural network model for the prediction of diseases and outcomes using patients' longitudinal EHRs. By first pretraining Enformer on a large collection of EHRs (255 million visits from 6.5 million patients between 2016 and 2019) and then finetuning for specific clinical applications, we found that Enformer outperformed existing SOTA models on a wide range of disease or outcome predictions. The performance gain was substantial on intentional self-harm prediction among PTSD patients. As shown in Table 3, Enformer was the best-performing model and outperformed BERT, LSTM, and logistic regression models by 24.3%, 96.9%, and 482.9%, respectively. The aforementioned results were not surprising, as deep-learning-based models are known to capture the salient information of EHR data to create powerful feature representations.[38] In addition, pretraining on large data and using the encoder-decoder framework have both been shown to be SOTA strategies for sequence-to-sequence applications.[22]

Enformer outperformed BERT for prediction of both single and multiple diseases/outcomes, as shown in Tables 2, 3, and Supplementary Table 2. In contrast to BERT, Enformer learned the representation of each clinical variable by predicting the complete diseases and outcomes with the help of cross-attention and decoder self-attention. Cross-attention allowed Enformer to selectively focus on sections of past visit ICD codes that were most relevant to predicting each ICD code in the future visit, and the decoder self-attention helped Enformer in predicting complete diseases and outcomes by leveraging already predicted primary diagnostic ICD codes to predict secondary codes that are less common. Thus, the performance gains were the highest among uncommon disease predictions. As shown in Table 2, compared with the BERT model, Enformer improved the Jaccard similarity score by an average of 15% in predicting 10 common diseases, and by an average of 45% in predicting 10 uncommon diseases. Enformer also substantially improved prediction for intentional self-harm prediction (AUPRC from 13.17 to 16.38, $p<0.001$) and pancreatic cancer prediction (AUPRC from 81.11 to 82.73, $p<0.001$).

Pretraining played a key role in improving the performance of our deep-learning-based models. To demonstrate, we chose intentional self-harm prediction and compared the performance between a BERT model pretrained on our pretraining cohort (as described in the Data section) and a BERT model with no pretraining (parameters were randomly initialized before fine-tuning). Pretrained BERT substantially outperformed the non-pretrained BERT (19.51% higher AUPRC and 29.55% higher PPV), as shown in Table 3. In particular, pretraining helps improve the latent representations of EHR data compared to non-pretrained models. This helped improve the probability distribution of candidate diseases or outcomes. As shown in Figure 3, after pretraining on a large EHR cohort, the probability distribution for the next visit ICD code changed from a random distribution to a learned representation of clinically relevant diseases or outcomes. While a pretrained model can capture the probability distribution at a large cohort level, finetuning it further can improve the performance for a specific application.

Attention-based models benefit from longitudinal EHRs with long histories (approximately 10 prior visits). As shown in Table 3, of the attention-based models (Enformer and BERT) finetuned on more than 5 visits, the AUPRC scores improved by 18% and 27%, respectively, in comparison with models finetuned on only 5 most recent visits. However, with the same experimental setup, the AUPRC scores of non-attention-based models such as LSTM and logistic regression decreased by 8% and 58%, respectively, when finetuned on more than 5 prior visits (on average 10) compared with the models finetuned on only

the 5 most recent visits. These findings were consistent with previous research,[39] which showed that LSTM, although good at mitigating the vanishing gradient challenge of RNN, remains suboptimal with long-time dependencies.

Our work is also related to predictive model studies focused on intentional self-harm.[40–44] Typically, these approaches sample thousands of patients and use probability tables, decision trees, and logistic regression to predict intentional self-harm. PPV plays a crucial role here in estimating the potential benefit of any intensive case management intervention, if one were to be implemented based on any such approach. PPV indicates the percentage of patients receiving an intervention who would otherwise attempt self-harm. Hartl et al.[42] predicted intentional self-harm among PTSD patients (prevalence ratio 5%, PPV 0%). Simon et al.[44] integrated EHR data and questionnaires for 2,960,929 patients to predict suicide attempts (prevalence ratio 1%) within 90 days of a mental health visit. The most successful model demonstrated a PPV of 5%. Enformer, on the other hand, achieved a PPV of 8.45% at 10% threshold for prediction of intentional self-harm among PTSD patients, which substantially outperformed the baselines (Table 3). In other words, out of the 100 highest-predicted-risk patients from 1,000 previously diagnosed PTSD patients, almost 8 patients would attempt intentional self-harm for the first time before the next visit. In comparison, for logistic regression the PPV at the 10% threshold was 2.17%. A practical suicide prevention tool must have a relatively high PPV to minimize the resource cost and intrusion directed at patients who will never attempt self-harm.[45] Therefore, Enformer may be a reliable and feasible option for designing an effective suicide prevention system.

In conclusion, our results have multiple clinical implications. First, existing predictive models tend to focus on single diseases. Yet the focus in clinical care, particularly in older people, is often on managing comorbidities.[16] A predictive model that can predict multiple diseases or outcomes may be useful in designing complex treatment plans. Second, a disease-specific approach could require building hundreds of different predictive models, an inefficient and costly use of resources whereas Enformer's unique pretraining objective enables it to predict all diseases and outcomes of a visit, immediately after the pretraining. Third, Enformer outperformed existing SOTA predictive models with a substantial performance gain, especially for uncommon disease/outcome predictions. Thus, our approach could assist in the development of screening algorithms to detect uncommon conditions. Fourth, Enformer can be easily generalized on an out-of-domain dataset with a significantly smaller training data. Enformer's strong transfer learning capability makes it a good fit for hospital settings with limited data and computing resources.

Despite the merits of this study, there are several limitations that provide opportunities for future improvements. First, we followed previous work on model pretraining and included only diagnostic ICD codes and demographic information.[12] However, other information such as procedure codes, medications, lab results, and phenotypical information extracted from notes can be added to further improve the performance.[46] Second, our prediction of single disease was limited to pancreatic cancer and single outcome was limited to intentional self-harm with PTSD. Future works would expand the set of diseases and outcomes.

Methods

Cohort and Use Case Definitions

VHA Cohort: Using the Veterans Health Administrations (VHA) Clinical Data Warehouse (CDW), we first identified a total of 8,308,742 patients who received care from more than 1,200 health care facilities of the US VHA from 1/1/2016 to 12/31/2019. For inclusion, we required each patient to have at least two visits: one outcome visit and at least one prior visit to be used for prediction of the outcome. This resulted in a total of 6,829,064 patients.

Pretraining and Finetuning Cohorts: Following the standard 95-5 split[12,21], we randomly took 95% of the patients (6,475,218) to create our pretraining cohort and used the remaining 5% (353,846) to create other datasets to finetune for DOAP and single disease or outcome predictions. The detailed patient cohort selections and diseases or outcomes for this study are shown in Figure 2 in the Supplement. The study protocol was approved by the Institutional Review Board at the VA Bedford Healthcare System.

We identified two use cases to evaluate Enformer.

Pancreatic Cancer: Although pancreatic cancer is relatively uncommon, it is deadly, projecting to become the second leading cause of cancer-related mortality by 2030.[29] Since effective screening is not available for pancreatic cancer, most patients can seek

medical attention only after being diagnosed
with locally advanced or metastatic cancer. Therefore, accurate prediction of pancreatic cancer can help
early detection of pancreatic cancer. Previous research[12]
has evaluated BERT for pancreatic cancer prediction. Following the same criteria in a previous study,[12] we defined pancreatic cancer with the first 3 digit ICD-10 codes as C25. 10,287 patients of 45 years or older with no report of any other cancer disease before their first pancreatic cancer diagnosis were included as cases, and 11,522 patients of 45 years or older without any cancer diagnosis were selected as controls.

Prediction of intentional self-harm: PTSD is considered as a hallmark injury of US Post-9/11 Veterans, with a prevalence estimated to be up to 23%.[47] Individuals with PTSD also have co-occurring physical health (e.g., chronic pain[48]), mental health (e.g., depression and anxiety[49,50]), and behavioral conditions (e.g., substance use disorder[51,52]). People with PTSD have 5.3-13 times the rate of suicide than people without PTSD.[53] Hence, we also evaluated Enformer for prediction of intentional self-harm for patients with PTSD. Out of 70,967 the patients who had been diagnosed with PTSD from 353,846 patients not in the pretraining cohort, 1,342 patients who would attempt self-harm for the first-time (from VHA CDW and Suicide Prevention Applications Network) were collected as cases. Following previous research that used the ICD-9 codes for intentional self-harm,[54] we converted those ICD-9 into ICD-10 and defined intentional self-harm using ICD-10 codes as in Supplementary Table 6. Another 281 patients were considered as controls who were diagnosed with PTSD but would not attempt self-harm in six months after their last visit. The observed self-harm rate was 1.9% among PTSD patients. Further details of the cohort definition are available in Supplementary Figure 3. All inpatient and outpatient ICD codes were included in our data.

Implementation Details

Experiments were conducted using Python version 3.8 and Transformer library version 4.16.2. For pretraining, we tasked Enformer to predict the next visit ICD codes recursively. Specifically, for each patient's input sequence, we used the first visit to predict the second visit, and then used the first two visits to predict the third visit, and so on. This process was repeated until the last visit as recorded in the input sequence was predicted. ICD codes were ordered by their priorities within a visit. We set the maximum sequence length (number of ICD codes) to be 512. To prevent the model from simply memorizing the pretraining data, we randomly drop 15% of visits from each input during pretraining. Other hyperparameters include a warmup ratio of 0.1, a learning rate of 1e−3, a dropout rate of 0.1, and an L2 weight decay of 1e−3 with fp16. To make a fair comparison, we pretrained the baseline BERT model on the same pretraining cohort, instead of using existing models pretrained on other cohorts. Other hyperparameters were the same as Enformer's. We used 4 Nvidia Tesla P40 GPU of 22 GB graphics memory capacity and trained each model for 6 days with more than 280k steps and batch size of 48.

During finetuning, we added a task-specific classifier layer (a linear layer) on top of Enformer and BERT to predict disease or outcome as a binary classification task. For a fair comparison with other medical BERT models,[12−14] Enformer used 6 layers in both encoder and decoder modules to best match the amount of parameters. All models were finetuned with 5 random seeds and statistical tests were carried out among these models. To ensure a fair comparison, the same feature transformation, L2 regularization, and hyperparameter tuning strategies, were used to finetune Enformer and all baseline models. For each outcome prediction, we built training, validation, and test datasets by the ratio of 7:1:2. One sided student's t-test was used to determine if Enformer outperforms baseline models. All finetuning experiments were conducted with Nvidia Tesla P40 GPU, and each single disease or outcome prediction was finetuned within 12 hours.

Declarations

Data Availability

The VHA EHR data cannot be shared publicly due to HIPAA regulation. MIMIC-IV is publicly available for the research community. The machine learning algorithms and software will be made available upon acceptance.

Competing Interests

The authors declare no competing interests.

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Tables

Table 1. Disease or outcome definitions in this study.

Task	Disease or Outcome (ICD-10 code)
Prediction of single disease or outcome	New onset pancreatic cancer (C25) Intentional self-harm among patients with PTSD
Disease/outcome agnostic prediction - common	Chronic post-traumatic stress disorder (F43.12) Type 2 diabetes (E11.9) Hyperlipidemia (E78.5) Loin pain (R10.3) Low back pain (M54.50) Obstructive sleep apnea (G47.33) Depression (F33.9) Obstructive airway disease (J44.9) Gastroesophageal reflux disease (K21.9) Arteriosclerosis (I25.10)
Disease/outcome agnostic prediction - uncommon	Benign neoplasm of connective tissue of eyelid (D21.0) Refractory anemia (D46.4) Melanocytic nevi of upper limb (D22.6) Benign neoplasm of skin of upper eyelid (D23.10) Cutaneous abscess of axilla (L02.41) Ankle and foot subacute osteomyelitis (M86.27) Cortical hemisphere nontraumatic hemorrhage intracerebral (I61.1) Malignant neoplasm of head of pancreas (C25.0) Other complication of kidney transplant (T86.19) Nonexudative age-related macular degeneration (H35.31)

Table 2. Performance of models for pancreatic cancer prediction. Result is calculated from best hyperparameters with 10 randomized seeds each.

		AUROC	AUPRC
		Mean (95% CI)	Mean (95% CI)
Without Pretraining	Logistic regression	77.71 (75.03-80.39)	63.81 (61.47-66.15)
	LSTM	81.34 (80.88-81.80)	78.46 (77.60-79.32)
	BERT without pretraining	81.35 (80.90-81.80)	78.65(77.87-79.43)
With Pretraining	BERT	83.42 (82.98-83.86)	81.11 (80.53-81.69)
	Enformer (ours)	85.35 (83.93-86.77)	82.73 (81.81-83.65)

Table 3. Performance (and standard deviation) of predictive models for intentional self-harm. Result is calculated from best hyperparameters with 10 randomized seeds each. "Self-Harm w/ Full History" refers to cases where the prediction is based on the original EHR (mean: 10.1 visits, st.dev.: 3.3 visits) prior to predicting intentional self-harm. "Self-Harm w/ Short History" includes only the 5 most recent visits.

	Models	Self-Harm w/ Short History			Self-Harm w/ Full History		
		AUPRC	PPV	F1	AUPRC	PPV	F1
		%	%	%	%	%	%
Without Pretraining	Logistic regression	6.89	3.37	6.28	2.81	2.96	5.67
		(1.57)	(0.04)	(0.07)	(1.16)	(0.87)	(1.61)
	LSTM	9.14	3.92	7.30	8.32	3.71	6.88
		(0.56)	(0.25)	(0.43)	(0.65)	(0.13)	(0.29)
	BERT without pretraining	9.42	5.41	8.99	11.02	6.1	9.28
		(0.62)	(0.46)	(0.87)	(0.75)	(1.29)	(0.77)
With Pretraining	BERT	10.31	5.78	9.49	13.17	7.14	10.47
		(0.89)	(1.50)	(1.78)	(0.72)	(1.11)	(0.99)
	Enformer	13.82	6.11	10.83	16.38	8.67	14.42
		(0.79)	(0.71)	(0.98)	(0.57)	(0.21)	(0.43)

Figures

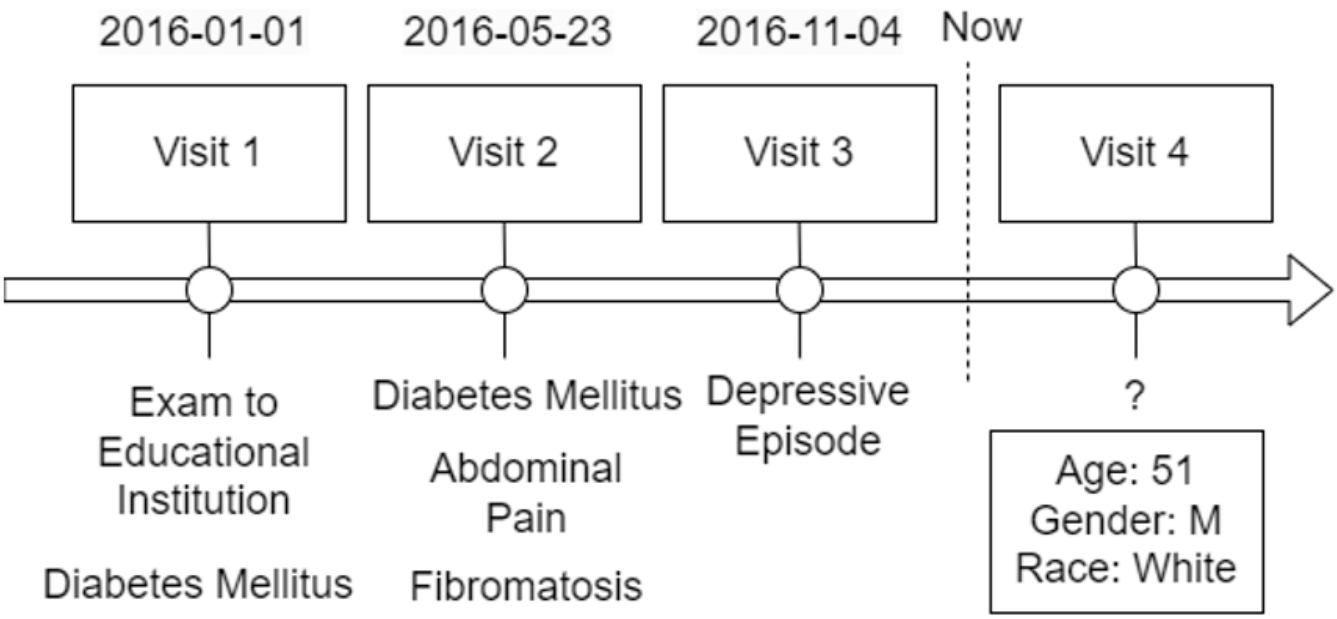

Figure 1

An example of patient's EHR sequence, containing demographic and ICD codes from 3 previous visits. The pretraining objective is to predict all diseases and outcomes in the next visit. All these values are artificial and for illustration purposes.

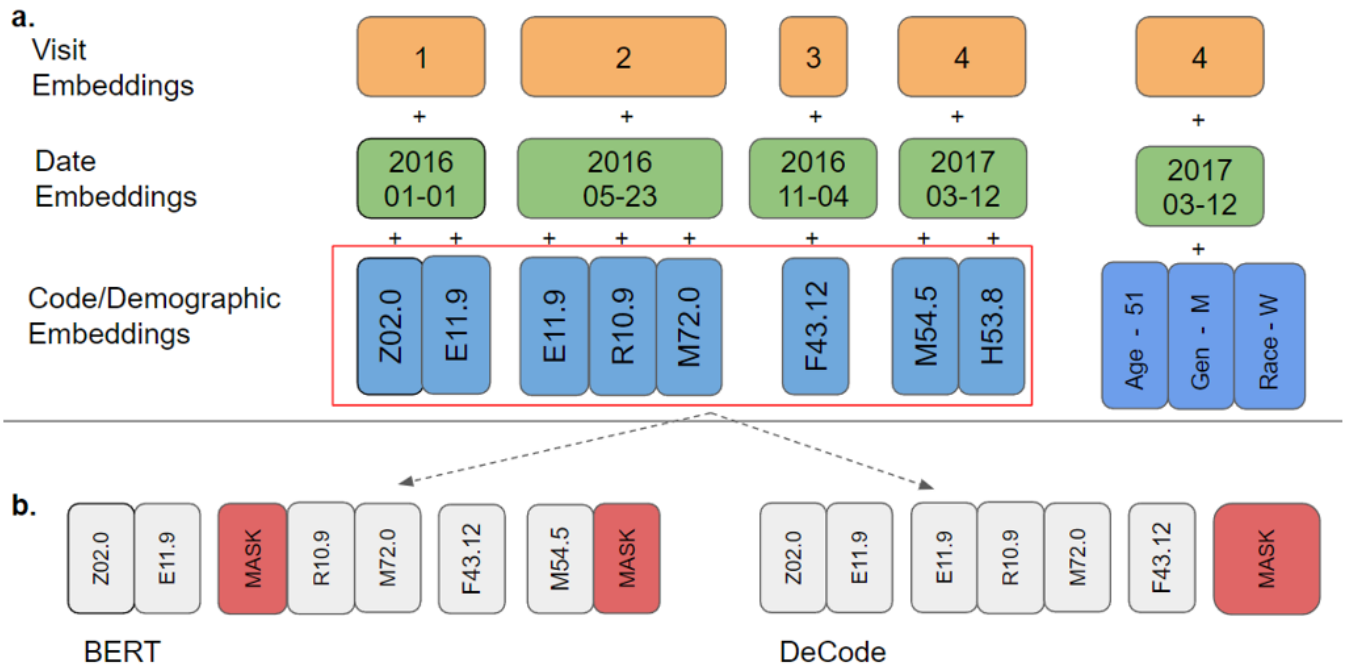

Figure 2

Input preprocessing from demographics and ICD codes from 4 visits shown in subfigure a and different medical history masking schemes in subfigure b. Our model learns the correlation of ICD codes by recovering the masked EHR to its original EHR.

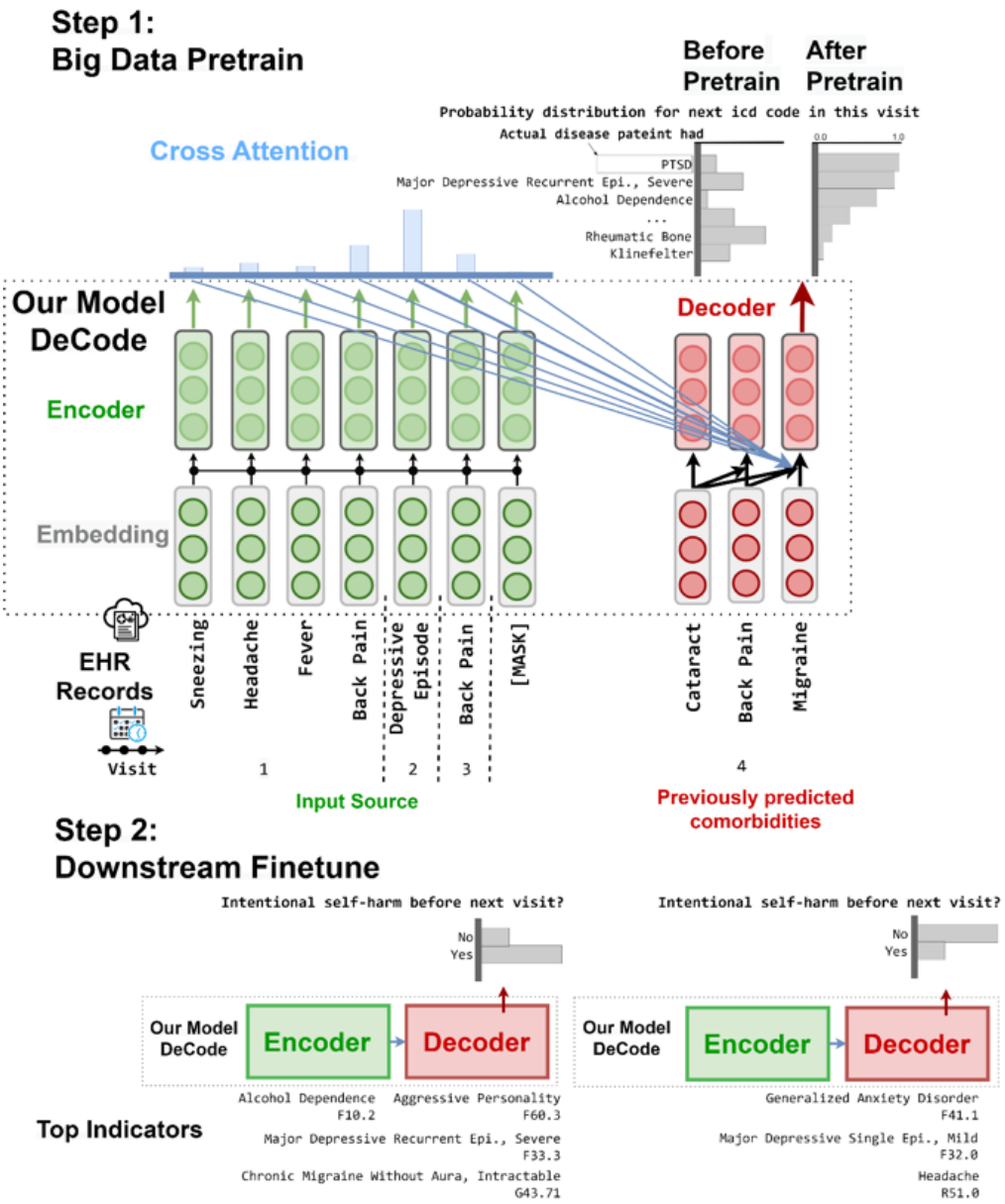

Figure 3

Enformer architecture and pretrain-finetune paradigm. Enformer was first pretrained with generative encoder-decoder transformer on a large set of longitudinal EHR data. During pretraining, Enformer learned the probability distribution of ICD codes (vs. random distribution) through correlation of cross attention. We then finetuned Enformer to the predictions of single disease or outcome. Through attention weights, Enformer was able to identify top indicators for the predictions.

Supplementary Files

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- Supplement.docx